BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 002 · Signed integers & binary arithmetic

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Choose a type

Q2. Keep the leading zeros

Q3. Read an unsigned byte

Q4. Can it fit?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — Sign magnitude

Recall the term — Two’s complement

Recall the term — Magnitude

Recall the term — Most significant bit

Recall the term — Signed overflow

Recall the term — Carry

Recall the term — Range

New learning

Sign magnitude — The leftmost bit stores the sign. The remaining bits store the positive magnitude. — CODE:
10000101
sign = 1
magnitude = 0000101 = 5
value = −5

Two’s complement — The most significant bit has a negative place value. — CODE:
11111011
= −128 + 64 + 32 + 16 + 8 + 2 + 1
= −5

Add using binary carries — CODE:
0 + 0 = 0
0 + 1 = 1
1 + 1 = 10
1 + 1 + 1 = 11 — Keep the fixed number of result bits. A carry beyond the width is discarded.

Overflow and subtraction — Signed overflow occurs when the mathematical result is outside the representable range. Two same-sign inputs producing an opposite-sign result is a warning sign.

Worked example

Encode −13 in two representations

1. Write +13 in eight bits — CODE:
13 = 8 + 4 + 1
+13 = 00001101 — Both signed representations begin from the positive magnitude.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q5. (a) 112 and 32 (both for 1).
(b) 10010000 (1).
(c) Correct mathematical total = 144 (1); 144 exceeds the eight-bit signed maximum of +127, so signed overflow occurs (1). The stored pattern represents −112 when interpreted as an 8-bit two’s-complement integer; it is not the correct mathematical sum.
(d) 9 bits (1), with signed range −256…255.
Indicative classroom marking guide; not an official OCR mark scheme. Q1 adapts OCR H446/01 June 2017 Q6(b)(i–ii). Remaining items are original exam-style practice.

Exit Ticket · Model answer

True. In two’s complement, eight bits represent values from −128 through +127.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Integer for the count; Boolean for open/not open.

Q2. String: the identifier must retain its leading zeros.

Q3. 128 + 32 + 4 + 2 = 166 .

Q4. 200 = 128 + 64 + 8 → 11001000 . Eight unsigned bits represent 0–255; 300 needs at least nine bits.

You Do pages · Independent questions

Q1. (a) +37 = 00100101 (1); sign magnitude −37 = 10100101 (1).
(b) Invert +37 → 11011010 (1); add 1 → 11011011 (1).

Q2. (a) Unsigned: 128 + 16 + 4 + 2 (1) = 150 (1).
(b) Sign magnitude: sign bit 1 with magnitude 16 + 4 + 2 = 22 (1) → −22 (1).
(c) Two’s complement: −128 + 16 + 4 + 2 (1) = −106 (1).

Q3. CODE:
00101101 (+45)
+ 00011011 (+27)

01001000 (+72)
Correct column addition with carries (1); 01001000 (1); 72 (1); no signed overflow (1). The total is within −128…127.
Starting on the right: column totals are 2, 2, 2, 3, 2, 2, 1, 0 when the incoming carries are included.

Q4. +29 = 00011101 (1). Invert to 11100010 , then add 1: −29 = 11100011 (1).
CODE:
00010010 (+18)
+ 11100011 (−29)

11110101
Correct addition of +18 and −29 (1); 11110101 (1); −128 + 64 + 32 + 16 + 4 + 1 = −11 (1).

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

2. Set the sign bit — CODE:
00001101
→ 10001101 — Keep the seven magnitude bits and change the leftmost sign bit to 1 .

3. Invert, then add 1 — CODE:
00001101
invert → 11110010
add 1 → 11110011

4. Convert the two’s-complement result to denary — CODE:
−128 + 64 + 32 + 16 + 2 + 1 = −13 — Invert-and-add-one negates a value in a fixed width.

We Do · Practise with guidance

2. Convert an 8-bit two’s-complement number to denary

Interpret the sign bit and show signed place values to find the denary integer.

Build the answer

Check: invert 11100110 → 00011001 ; add 1 → 00011010 = 26.

3. Add 23 and 19

Use the method from I Do to build the answer.

4. Subtract 19 from 12

19: 00010011 → invert 11101100 → add 1 11101101 .

5. Spot signed overflow

Two positive inputs produced a negative stored result.

Represent −44

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | You Do Work independently | Review & Improve Check my understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Two representations
Represent −37 in (a) sign magnitude [2] and (b) two’s complement [2]. Show the positive binary value and the change made for each method.

Question 2

The same pattern
The pattern is 10010110 . Determine its denary value when interpreted as (a) unsigned binary, (b) sign magnitude and (c) two’s complement. Show working. [2 each]

Question 3

Signed binary addition
Add 00101101 and 00011011 as two’s-complement integers. Show column working or carries [1], the stored result [1], its denary value [1], and whether signed overflow occurs [1].

Question 4

Subtract using two’s complement
Calculate 18 − 29 using eight-bit two’s complement. Show +29 [1], −29 [1], the binary addition [1], the stored result [1] and its signed denary value [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Question 5

Explain an overflow
A processor adds 01110000 and 00100000 as eight-bit two’s-complement integers.
State both input values in denary. [1]
Give the stored eight-bit result. [1]
Explain whether signed overflow has occurred. Refer to the mathematical total and the representable range. [2]
State the minimum signed two’s-complement width needed to store the correct result. [1]

Exit Ticket · Answer before checking

In 8-bit two’s complement, the signed range is −128 to +127.